IN-BEAM GAMMA-RAY SPECTROSCOPY

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Appendix: Table of Angular Distribution Functions

Table of Angular-Distribution Functions

In this appendix numerical values of the angular distribution functions are given for quick use of the formulae derived in this chapter. Table 2A.1 gives the statistical tensors B_k, equation (2.48), in the limiting case of perfect alignment. In table 2.A2 F_k values are listed, as given by equation (2.42), and B_kF_k values which give the angular-distribution coefficients of a perfectly aligned state. The angular-distribution is then expressed by equation (2.43) or equation (2.49). The U_k coefficients are also given. These tables were prepared by Yamazaki [1967].

BK	=	$B_k(J) = \rho_k(J)$
JI	=	J_i Initial spin
JF	=	J_f Final spin
L1	=	λ_1 } Angular momenta of a mixed transition
L2	=	λ_2
F2	=	$F_2(J_f\lambda_1\lambda_2J_i)$
F4	=	$F_4(J_f\lambda_1\lambda_2J_i)$
B2F2	=	$B_2(J_i)F_2(J_f\lambda_1\lambda_2J_i)$
B4F4	=	$B_4(J_i)F_4(J_f\lambda_1\lambda_2J_i)$
U2	=	$u_2(J_i\lambda J_f)$
U4	=	$u_4(J_i\lambda J_f)$

Table 2A.1

Statistical tensors for complete alignment, B_k (J)

J	B2	B4	B6
1	−1.41421	0.	0.
2	−1.19523	1.60357	0.
3	−1.15470	1.27920	−1.74078
4	−1.13961	1.20687	−1.34840
5	−1.13228	1.17670	−1.25245
6	−1.12815	1.16086	−1.20977
7	−1.12560	1.15142	−1.18628
8	−1.12390	1.14531	−1.17175
9	−1.12272	1.14112	−1.16206
10	−1.12187	1.13811	−1.15524
11	−1.12122	1.13588	−1.15025
12	−1.12073	1.13418	−1.14648
13	−1.12034	1.13285	−1.14356
14	−1.12004	1.13179	−1.14125
15	−1.11979	1.13093	−1.13939
16	−1.11958	1.13022	−1.13786
17	−1.11941	1.12964	−1.13660
18	−1.11926	1.12915	−1.13554
19	−1.11914	1.12873	−1.13465
20	−1.11903	1.12837	−1.13388
3/2	−1.00000	0.	0.
5/2	−1.06904	.92582	0.
7/2	−1.09109	1.02565	−.87039
9/2	−1.10096	1.06436	−.98473
11/2	−1.10624	1.08389	−1.03418
13/2	−1.10940	1.09522	−1.06104
15/2	−1.11144	1.10240	−1.07749
17/2	−1.11283	1.10725	−1.08837
19/2	−1.11382	1.11068	−1.09596
21/2	−1.11456	1.11321	−1,10148
23/2	−1.11511	1.11511	−1.10563
25/2	−1.11555	1.11659	−1.10882
27/2	−1.11589	1.11776	−1.11134
29/2	−1.11617	1.11870	−1.11336
31/2	−1.11639	1.11947	−1.11501
33/2	−1.11658	1.12010	−1.11636
35/2	−1.11674	1.12064	−1.11750
37/2	−1.11687	1.12109	−1.11845
39/2	−1.11698	1.12147	−1.11927
41/2	−1.11708	1.12180	−1.11997

Table 2A.2. Angular distribution functions.

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
1	0	1	1	0.7071	-1.0000	0.0	0.0	0.0	0.0
1	1	1	1	-0.3536	0.5000	0.0	0.0	-0.5000	0.0
1	1	1	2	-1.0607	1.5000	0.0	0.0	0.0	0.0
1	1	2	2	-0.3536	0.5000	0.0	0.0	0.1000	0.0
1	2	1	1	0.0707	-0.1000	0.0	0.0	0.5916	0.0
1	2	1	2	0.4743	-0.6708	0.0	0.0	0.0	0.0
1	2	2	2	0.3536	-0.5000	0.0	0.0	-0.5916	0.0
1	3	2	2	-0.1010	0.1429	0.0	0.0	0.4899	0.0
1	3	2	3	0.3780	-0.5345	0.0	0.0	0.0	0.0
1	3	3	3	0.5303	-0.7500	0.0	0.0	-0.6124	0.0
1	4	3	3	-0.1768	0.2500	0.0	0.0	0.4432	0.0
2	0	2	2	-0.5976	0.7143	-1.0590	-1.7143	0.0	0.0
2	1	1	1	0.4183	-0.5000	0.0	0.0	0.5916	0.0
2	1	1	2	-0.9354	1.1180	0.0	0.0	0.0	0.0
2	1	2	2	-0.2988	0.3571	0.7127	1.1429	-0.5916	0.0
2	2	1	1	-0.4183	0.5000	0.0	0.0	0.5000	-0.6667
2	2	1	2	-0.6124	0.7319	0.0	0.0	0.0	0.0
2	2	2	2	0.1281	-0.1531	-0.3054	-0.4898	-0.2143	0.2857
2	3	1	1	0.1195	-0.1429	0.0	0.0	0.8281	0.4179
2	3	1	2	0.6547	-0.7825	0.0	0.0	0.0	0.0
2	3	2	2	0.3415	-0.4082	0.0764	0.1224	0.2070	-0.6268
2	4	2	2	-0.1707	0.2041	-0.0085	-0.0136	0.7491	0.2847
2	4	2	3	0.5051	-0.6037	-0.0627	-0.1006	0.0	0.0
2	4	3	3	0.4482	-0.5357	-0.0297	-0.0476	0.0749	-0.5694
2	5	3	3	-0.2988	0.3571	0.0040	0.0065	0.7037	0.2271
3	0	3	3	-0.8660	1.0000	0.2132	0.2727	0.0	0.0
3	1	2	2	-0.4949	0.5714	-0.4467	-0.5714	0.4899	0.0
3	1	2	3	-0.4629	0.5345	1.0446	1.3363	0.0	0.0
3	1	3	3	-0.6495	0.7500	0.0355	0.0455	-0.6124	0.0
3	2	1	1	0.3464	-0.4000	0.0	0.0	0.8281	0.4179

JT	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
3	2	1	2	-0.9487	1.0954	0.0	0.0	0.0	0.0
3	2	2	2	-0.1237	0.1429	0.6701	0.8571	0.2070	-0.6268
3	3	1	1	-0.4330	0.5000	0.0	0.0	0.7500	0.1667
3	3	1	2	-0.4330	0.5000	0.0	0.0	0.0	0.0
3	3	2	2	0.2268	-0.2619	-0.4467	-0.5714	0.3167	-0.5000
3	4	1	1	0.1443	-0.1667	0.0	0.0	0.9047	0.6814
3	4	1	2	0.7217	-0.8333	0.0	0.0	0.0	0.0
3	4	2	2	0.3093	-0.3571	0.1489	0.1905	0.5428	-0.2271
3	5	2	2	-0.2062	0.2381	-0.0203	-0.0260	0.8498	0.5436
3	5	2	3	0.5455	-0.6299	-0.1343	-0.1718	0.0	0.0
3	5	3	3	0.3608	-0.4167	-0.0549	-0.0702	0.4249	-0.3624
3	6	3	3	-0.3608	0.4167	0.0097	0.0124	0.8142	0.4675
4	1	3	3	-0.7835	0.8929	0.1453	0.1753	0.4432	0.0
4	2	2	2	-0.4477	0.5102	-0.3044	-0.3673	0.7491	0.2847
4	2	2	3	-0.5297	0.6037	0.9004	1.0866	0.0	0.0
4	2	3	3	-0.4701	0.5357	-0.0484	-0.0584	0.0749	-0.5694
4	3	1	1	0.3134	-0.3571	0.0	0.0	0.9047	0.6814
4	3	1	2	-0.9402	1.0714	0.0	0.0	0.0	0.0
4	3	2	2	-0.0448	0.0510	0.6088	0.7347	0.5428	-0.2271
4	4	1	1	-0.4387	0.5000	0.0	0.0	0.8500	0.5000
4	4	1	2	-0.3354	0.3822	0.0	0.0	0.0	0.0
4	4	2	2	0.2646	-0.3015	-0.4981	-0.6011	0.5734	-0.1494
4	5	1	1	0.1595	-0.1818	0.0	0.0	0.9394	0.7977
4	5	1	2	0.7568	-0.8624	0.0	0.0	0.0	0.0
4	5	2	2	0.2849	-0.3247	0.1937	0.2338	0.7045	0.1330
4	6	2	2	-0.2279	0.2597	-0.0298	-0.0360	0.9000	0.6861
4	6	2	3	0.5641	-0.6428	-0.1844	-0.2225	0.0	0.0
4	6	3	3	0.2991	-0.3409	-0.0687	-0.0830	0.6107	-0.0490
4	7	3	3	-0.3989	0.4545	0.0142	0.0172	0.8723	0.6157
5	2	3	3	-0.7360	0.8333	0.1159	0.1364	0.7037	0.2271

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
5	3	2	2	-0.4206	0.4762	-0.2428	-0.2857	0.8498	0.5436
5	3	2	3	-0.5563	0.6299	0.8030	0.9449	0.0	0.0
5	3	3	3	-0.3680	0.4167	-0.0773	-0.0909	0.4249	-0.3624
5	4	1	1	0.2944	-0.3333	0.0	0.0	0.9394	0.7977
5	4	1	2	-0.9309	1.0541	0.0	0.0	0.0	0.0
5	4	2	2	0.0	0.0	0.5666	0.6667	0.7045	0.1330
5	5	1	1	-0.4416	0.5000	0.0	0.0	0.9000	0.6667
5	5	1	2	-0.2739	0.3101	0.0	0.0	0.0	0.0
5	5	2	2	0.2831	-0.3205	-0.5230	-0.6154	0.7103	0.1538
5	6	1	1	0.1698	-0.1923	0.0	0.0	0.9580	0.8601
5	6	1	2	0.7783	-0.8813	0.0	0.0	0.0	0.0
5	6	2	2	0.2669	-0.3022	0.2241	0.2637	0.7938	0.3686
5	7	2	2	-0.2426	0.2747	-0.0374	-0.0440	0.9286	0.7718
5	7	2	3	0.5742	-0.6501	-0.2210	-0.2600	0.0	0.0
5	7	3	3	0.2548	-0.2885	-0.0773	-0.0909	0.7196	0.1930
5	8	3	3	-0.4246	0.4808	0.0178	0.0210	0.9067	0.7113
6	3	3	3	-0.7051	0.7955	0.0997	0.1157	0.8142	0.4675
6	4	2	2	-0.4029	0.4545	-0.2088	-0.2424	0.9000	0.6861
6	4	2	3	-0.5698	0.6428	0.7383	0.8571	0.0	0.0
6	4	3	3	-0.3022	0.3409	-0.0902	-0.1047	0.6107	-0.0490
6	5	1	1	0.2820	-0.3182	0.0	0.0	0.9580	0.8601
6	5	1	2	-0.9232	1.0415	0.0	0.0	0.0	0.0
6	5	2	2	0.0288	-0.0325	0.5370	0.6234	0.7938	0.3686
6	6	1	1	-0.4432	0.5000	0.0	0.0	0.9286	0.7619
6	6	1	2	-0.2315	0.2611	0.0	0.0	0.0	0.0
6	6	2	2	0.2935	-0.3312	-0.5370	-0.6234	0.7909	0.3636
6	7	1	1	0.1773	-0.2000	0.0	0.0	0.9692	0.8974
6	7	1	2	0.7928	-0.8944	0.0	0.0	0.0	0.0
6	7	2	2	0.2533	-0.2857	0.2461	0.2857	0.8481	0.5235
6	8	2	2	-0.2533	0.2857	-0.0434	-0.0504	0.9464	0.8270

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
6	8	2	3	0.5803	-0.6547	-0.2488	-0.2888	0.0	0.0
6	8	3	3	0.2216	-0.2500	-0.0829	-0.0963	0.7887	0.3676
6	9	3	3	-0.4432	0.5000	0.0207	0.0241	0.9288	0.7758
7	4	3	3	-0.6834	0.7692	0.0894	0.1030	0.8723	0.6157
7	5	2	2	-0.3905	0.4396	-0.1874	-0.2158	0.9286	0.7718
7	5	2	3	-0.5776	0.6501	0.6929	0.7979	0.0	0.0
7	5	3	3	-0.2563	0.2885	-0.0969	-0.1116	0.7196	0.1930
7	6	1	1	0.2734	-0.3077	0.0	0.0	0.9692	0.8974
7	6	1	2	-0.9169	1.0320	0.0	0.0	0.0	0.0
7	6	2	2	0.0488	-0.0549	0.5154	0.5934	0.8481	0.5235
7	7	1	1	-0.4442	0.5000	0.0	0.0	0.9464	0.8214
7	7	1	2	-0.2004	0.2256	0.0	0.0	0.0	0.0
7	7	2	2	0.3001	-0.3378	-0.5457	-0.6283	0.8422	0.5079
7	8	1	1	0.1829	-0.2059	0.0	0.0	0.9765	0.9216
7	8	1	2	0.8033	-0.9041	0.0	0.0	0.0	0.0
7	9	2	2	0.2426	-0.2731	0.2527	0.3025	0.8835	0.6290
7	9	2	2	-0.2613	0.2941	-0.0484	-0.0557	0.9583	0.8645
7	9	2	3	0.5843	-0.6577	-0.2706	-0.3115	0.0	0.0
7	9	3	3	0.1960	-0.2206	-0.0869	-0.1001	0.8351	0.4940
7	10	3	3	-0.4573	0.5147	0.0231	0.0266	0.9439	0.8211
8	5	3	3	-0.6673	0.7500	0.0824	0.0944	0.9067	0.7113
8	6	2	2	-0.3813	0.4286	-0.1727	-0.1978	0.9464	0.8270
8	6	2	3	-0.5825	0.6547	0.6595	0.7554	0.0	0.0
8	6	3	3	-0.2224	0.2500	-0.1007	-0.1154	0.7887	0.3676
8	7	1	1	0.2669	-0.3000	0.0	0.0	0.9765	0.9216
8	7	1	2	-0.9117	1.0247	0.0	0.0	0.0	0.0
8	7	2	2	0.0536	-0.0714	0.4989	0.5714	0.8835	0.6290
8	8	1	1	-0.4449	0.5000	0.0	0.0	0.9583	0.8611
8	8	1	2	-0.1768	0.1987	0.0	0.0	0.0	0.0
8	8	2	2	0.3044	-0.3421	-0.5514	-0.6316	0.8768	0.6097

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
8	9	1	1	0.1873	-0.2105	0.0	0.0	0.9814	0.9381
8	9	1	2	0.8111	-0.9116	0.0	0.0	0.0	0.0
8	9	2	2	0.2341	-0.2632	0.2757	0.3158	0.9078	0.7036
8	10	2	2	-0.2676	0.3008	-0.0525	-0.0601	0.9667	0.8910
8	10	2	3	0.5870	-0.6598	-0.2880	-0.3299	0.0	0.0
8	10	3	3	0.1756	-0.1974	-0.0898	-0.1029	0.8678	0.5873
8	11	3	3	-0.4683	0.5263	0.0251	0.0287	0.9547	0.8541
9	6	3	3	-0.6549	0.7353	0.0773	0.0882	0.9288	0.7758
9	7	2	2	-0.3742	0.4202	-0.1620	-0.1849	0.9583	0.8645
9	7	2	3	-0.5858	0.6577	0.6340	0.7234	0.0	0.0
9	7	3	3	-0.1965	0.2206	-0.1031	-0.1176	0.8351	0.4940
9	8	1	1	0.2620	-0.2941	0.0	0.0	0.9814	0.9381
9	8	1	2	-0.9075	1.0188	0.0	0.0	0.0	0.0
9	8	2	2	0.0748	-0.0840	0.4860	0.5546	0.9078	0.7036
9	9	1	1	-0.4453	0.5000	0.0	0.0	0.9667	0.8889
9	9	1	2	-0.1581	0.1775	0.0	0.0	0.0	0.0
9	9	2	2	0.3074	-0.3451	-0.5555	-0.6338	0.9311	0.6835
9	10	1	1	0.1909	-0.2143	0.0	0.0	0.9850	0.9499
9	10	1	2	0.8172	-0.9175	0.0	0.0	0.0	0.0
9	10	2	2	0.2272	-0.2551	0.2861	0.3265	0.9253	0.7580
9	11	2	2	-0.2727	0.3061	-0.0560	-0.0639	0.9727	0.9105
9	11	2	3	0.5890	-0.6613	-0.3024	-0.3450	0.0	0.0
9	11	3	3	0.1590	-0.1786	-0.0920	-0.1050	0.8917	0.6576
9	12	3	3	-0.4772	0.5357	0.0267	0.0305	0.9626	0.8789
10	7	3	3	-0.6451	0.7237	0.0734	0.0836	0.9439	0.8211
10	8	2	2	-0.3686	0.4135	-0.1539	-0.1751	0.9667	0.8910
10	8	2	3	-0.5881	0.6598	0.6138	0.6986	0.0	0.0
10	8	3	3	-0.1759	0.1974	-0.1046	-0.1191	0.8678	0.5873
10	9	1	1	0.2580	-0.2895	0.0	0.0	0.9850	0.9499
10	9	1	2	-0.9039	1.0141	0.0	0.0	0.0	0.0

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
10	9	2	2	0.0838	-0.0940	0.4757	0.5413	0.9253	0.7580
10	10	1	1	-0.4457	0.5000	0.0	0.0	0.9727	0.9091
10	10	1	2	-0.1430	0.1604	0.0	0.0	0.0	0.0
10	10	2	2	0.3096	-0.3473	-0.5584	-0.6355	0.9189	0.7385
10	11	1	1	0.1938	-0.2174	0.0	0.0	0.9876	0.9586
10	11	1	2	0.8221	-0.9223	0.0	0.0	0.0	0.0
10	11	2	2	0.2215	-0.2484	0.2947	0.3354	0.9382	0.7988
10	12	2	2	-0.2768	0.3106	-0.0589	-0.0671	0.9773	0.9252
10	12	2	3	0.5905	-0.6624	-0.3143	-0.3577	0.0	0.0
10	12	3	3	0.1453	-0.1630	-0.0938	-0.1067	0.9096	0.7117
10	13	3	3	-0.4844	0.5435	0.0281	0.0320	0.9686	0.8978
11	8	3	3	-0.6371	0.7143	0.0704	0.0800	0.9547	0.8541
11	9	2	2	-0.3640	0.4082	-0.1475	-0.1676	0.9727	0.9105
11	9	2	3	-0.5898	0.6613	0.5975	0.6787	0.0	0.0
11	9	3	3	-0.1593	0.1786	-0.1056	-0.1200	0.8917	0.6576
11	10	1	1	0.2548	-0.2857	0.0	0.0	0.9876	0.9586
11	10	1	2	-0.9009	1.0101	0.0	0.0	0.0	0.0
11	10	2	2	0.0910	-0.1020	0.4671	0.5306	0.9382	0.7988
11	11	1	1	-0.4459	0.5000	0.0	0.0	0.9773	0.9242
11	11	1	2	-0.1306	0.1464	0.0	0.0	0.0	0.0
11	11	2	2	0.3112	-0.3490	-0.5606	-0.6367	0.9323	0.7805
11	12	1	1	0.1962	-0.2200	0.0	0.0	0.9896	0.9652
11	12	1	2	0.8261	-0.9263	0.0	0.0	0.0	0.0
11	12	2	2	0.2166	-0.2429	0.3018	0.3429	0.9480	0.8302
11	13	2	2	-0.2803	0.3143	-0.0615	-0.0698	0.9808	0.9366
11	13	2	3	0.5916	-0.6633	-0.3244	-0.3685	0.0	0.0
11	13	3	3	0.1338	-0.1500	-0.0952	-0.1081	0.9235	0.7562
11	14	3	3	-0.4905	0.5500	0.0293	0.0333	0.9733	0.9127
12	9	3	3	-0.6304	0.7065	0.0680	0.0771	0.9626	0.8789
12	10	2	2	-0.3602	0.4037	-0.1424	-0.1615	0.9773	0.9252

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
12	10	2	3	-0.5911	0.6624	0.5841	0.6624	0.0	0.0
12	10	3	3	-0.1455	0.1630	-0.1063	-0.1206	0.9096	0.7117
12	11	1	1	0.2522	-0.2826	0.0	0.0	0.9896	0.9652
12	11	1	2	-0.8984	1.0068	0.0	0.0	0.0	0.0
12	11	2	2	0.0970	-0.1087	0.4500	0.5217	0.9480	0.8302
12	12	1	1	-0.4461	0.5000	0.0	0.0	0.9808	0.9359
12	12	1	2	-0.1201	0.1346	0.0	0.0	0.0	0.0
12	12	2	2	0.3125	-0.3502	-0.5622	-0.6377	0.9427	0.8133
12	13	1	1	0.1983	-0.2222	0.0	0.0	0.9911	0.9704
12	13	1	2	0.8295	-0.9296	0.0	0.0	0.0	0.0
12	13	2	2	0.2124	-0.2381	0.3079	0.3492	0.9557	0.8549
12	14	2	2	-0.2833	0.3175	-0.0637	-0.0723	0.9835	0.9456
12	14	2	3	0.5925	-0.6640	-0.3331	-0.3778	0.0	0.0
12	14	3	3	0.1239	-0.1389	-0.0963	-0.1092	0.9343	0.7880
12	15	3	3	-0.4957	0.5556	0.0304	0.0345	0.9770	0.9246
13	10	3	3	-0.6248	0.7000	0.0659	0.0747	0.9686	0.8978
13	11	2	2	-0.3570	0.4000	-0.1382	-0.1565	0.9808	0.9366
13	11	2	3	-0.5921	0.6633	0.5728	0.6489	0.0	0.0
13	11	3	3	-0.1339	0.1500	-0.1068	-0.1209	0.9235	0.7542
13	12	1	1	0.2499	-0.2800	0.0	0.0	0.9911	0.9704
13	12	1	2	-0.8961	1.0040	0.0	0.0	0.0	0.0
13	12	2	2	0.1020	-0.1143	0.4540	0.5143	0.9557	0.8549
13	13	1	1	-0.4463	0.5000	0.0	0.0	0.9835	0.9451
13	13	1	2	-0.1112	0.1246	0.0	0.0	0.0	0.0
13	13	2	2	0.3135	-0.3512	-0.5636	-0.6384	0.9508	0.8393
13	14	1	1	0.2001	-0.2241	0.0	0.0	0.9923	0.9745
13	14	1	2	0.8323	-0.9325	0.0	0.0	0.0	0.0
13	14	2	2	0.2089	-0.2340	0.3131	0.3547	0.9618	0.8745
13	15	2	2	-0.2858	0.3202	-0.0656	-0.0744	0.9857	0.9528
13	15	2	3	0.5932	-0.6646	-0.3406	-0.3859	0.0	0.0

JT	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
13	15	3	3	0.1154	-0.1293	-0.0972	-0.1101	0.9431	0.8154
13	16	3	3	-0.5002	0.5603	0.0313	0.0355	0.9799	0.9342
14	11	3	3	-0.6200	0.6944	0.0643	0.0727	0.9733	0.9127
14	12	2	2	-0.3543	0.3968	-0.1346	-0.1524	0.9835	0.9456
14	12	2	3	-0.5929	0.6640	0.5632	0.6375	0.0	0.0
14	12	3	3	-0.1240	0.1389	-0.1071	-0.1212	0.9343	0.7880
14	13	1	1	0.2480	-0.2779	0.0	0.0	0.9923	0.9745
14	13	1	2	-0.8942	1.0015	0.0	0.0	0.0	0.0
14	13	2	2	0.1063	-0.1190	0.4488	0.5079	0.9618	0.8745
14	14	1	1	-0.4464	0.5000	0.0	0.0	0.9857	0.9524
14	14	1	2	-0.1035	0.1159	0.0	0.0	0.0	0.0
14	14	2	2	0.3143	-0.3520	-0.5546	-0.6390	0.9573	0.8602
14	15	1	1	0.2016	-0.2258	0.0	0.0	0.9933	0.9777
14	15	1	2	0.8347	-0.9349	0.0	0.0	0.0	0.0
14	15	2	2	0.2057	-0.2304	0.3176	0.3595	0.9667	0.8905
14	16	2	2	-0.2880	0.3226	-0.0674	-0.0762	0.9875	0.9586
14	16	2	3	0.5937	-0.6650	-0.3472	-0.3930	0.0	0.0
14	16	3	3	0.1080	-0.1210	-0.0980	-0.1109	0.9502	0.8378
14	17	3	3	-0.5040	0.5645	0.0322	0.0364	0.9824	0.9420
15	12	3	3	-0.6159	0.6897	0.0628	0.0711	0.9770	0.9246
15	13	2	2	-0.3519	0.3941	-0.1316	-0.1489	0.9857	0.9528
15	13	2	3	-0.5935	0.6646	0.5550	0.6276	0.0	0.0
15	13	3	3	-0.1155	0.1293	-0.1073	-0.1214	0.9431	0.8154
15	14	1	1	0.2464	-0.2759	0.0	0.0	0.9933	0.9777
15	14	1	2	-0.8925	0.9994	0.0	0.0	0.0	0.0
15	14	2	2	0.1100	-0.1232	0.4443	0.5025	0.9667	0.8905
15	15	1	1	-0.4465	0.5000	0.0	0.0	0.9875	0.9583
15	15	1	2	-0.0968	0.1084	0.0	0.0	0.0	0.0
15	15	2	2	0.3149	-0.3527	-0.5655	-0.6395	0.9627	0.8773
15	16	1	1	0.2030	-0.2273	0.0	0.0	0.9941	0.9804

JT	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
15	16	1	2	0.8368	-0.9371	0.0	0.0	0.0	0.0
15	16	2	2	0.2030	-0.2273	0.3215	0.3636	0.9707	0.9035
15	17	2	2	-0.2899	0.3247	-0.0689	-0.0779	0.9890	0.9635
15	17	2	3	0.5942	-0.6654	-0.3530	-0.3992	0.0	0.0
15	17	3	3	0.1015	-0.1136	-0.0987	-0.1116	0.9560	0.8564
15	18	3	3	-0.5074	0.5682	0.0329	0.0372	0.9844	0.9486
16	13	3	3	-0.6123	0.6855	0.0616	0.0696	0.9799	0.9342
16	14	2	2	-0.3499	0.3917	-0.1291	-0.1459	0.9875	0.9586
16	14	2	3	-0.5940	0.6650	0.5478	0.6192	0.0	0.0
16	14	3	3	-0.1080	0.1210	-0.1075	-0.1215	0.9502	0.8378
16	15	1	1	0.2449	-0.2742	0.0	0.0	0.9941	0.9804
16	15	1	2	-0.8910	0.9975	0.0	0.0	0.0	0.0
16	15	2	2	0.1132	-0.1267	0.4403	0.4977	0.9707	0.9035
16	16	1	1	-0.4466	0.5000	0.0	0.0	0.9890	0.9632
16	16	1	2	-0.0909	0.1018	0.0	0.0	0.0	0.0
16	16	2	2	0.3155	-0.3532	-0.5562	-0.6399	0.9670	0.8915
16	17	1	1	0.2042	-0.2286	0.0	0.0	0.9948	0.9827
16	17	1	2	0.8387	-0.9389	0.0	0.0	0.0	0.0
16	17	2	2	0.2005	-0.2245	0.3250	0.3674	0.9741	0.9144
16	18	2	2	-0.2917	0.3265	-0.0703	-0.0794	0.9902	0.9675
16	18	2	3	0.5946	-0.6657	-0.3582	-0.4048	0.0	0.0
16	18	3	3	0.0957	-0.1071	-0.0992	-0.1121	0.9609	0.8720
16	19	3	3	-0.5104	0.5714	0.0335	0.0379	0.9861	0.9541
17	14	3	3	-0.6091	0.6818	0.0605	0.0684	0.9824	0.9420
17	15	2	2	-0.3480	0.3896	-0.1268	-0.1433	0.9890	0.9635
17	15	2	3	-0.5944	0.6654	0.5415	0.6117	0.0	0.0
17	15	3	3	-0.1015	0.1136	-0.1076	-0.1216	0.9560	0.8564
17	16	1	1	0.2436	-0.2727	0.0	0.0	0.9948	0.9827
17	16	1	2	-0.8896	0.9958	0.0	0.0	0.0	0.0
17	16	2	2	0.1160	-0.1299	0.4369	0.4935	0.9741	0.9144

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
17	17	1	1	-0.4467	0.5000	0.0	0.0	0.9902	0.9673
17	17	1	2	-0.0858	0.0950	0.0	0.0	0.0	0.0
17	17	2	2	0.3159	-0.3536	-0.5668	-0.6402	0.9707	0.9034
17	18	1	1	0.2052	-0.2297	0.0	0.0	0.9954	0.9846
17	18	1	2	0.8403	-0.9406	0.0	0.0	0.0	0.0
17	18	2	2	0.1983	-0.2220	0.3281	0.3707	0.9769	0.9236
17	19	2	2	-0.2932	0.3282	-0.0715	-0.0808	0.9912	0.9709
17	19	2	3	0.5949	-0.6660	-0.3628	-0.4098	0.0	0.0
17	19	3	3	0.0905	-0.1014	-0.0997	-0.1126	0.9650	0.8853
17	20	3	3	-0.5131	0.5743	0.0341	0.0386	0.9875	0.9588
18	15	3	3	-0.6063	0.6786	0.0596	0.0673	0.9844	0.9486
18	16	2	2	-0.3464	0.3878	-0.1249	-0.1410	0.9902	0.9675
18	16	2	3	-0.5948	0.6657	0.5360	0.6052	0.0	0.0
18	16	3	3	-0.0957	0.1071	-0.1077	-0.1216	0.9609	0.8720
18	17	1	1	0.2425	-0.2714	0.0	0.0	0.9954	0.9846
18	17	1	2	-0.8884	0.9944	0.0	0.0	0.0	0.0
18	17	2	2	0.1185	-0.1327	0.4338	0.4898	0.9769	0.9236
18	18	1	1	-0.4467	0.5000	0.0	0.0	0.9912	0.9708
18	18	1	2	-0.0811	0.0908	0.0	0.0	0.0	0.0
18	18	2	2	0.3163	-0.3540	-0.5573	-0.6405	0.9738	0.9134
18	19	1	1	0.2062	-0.2309	0.0	0.0	0.9958	0.9861
18	19	1	2	0.8417	-0.9421	0.0	0.0	0.0	0.0
18	19	2	2	0.1964	-0.2198	0.3309	0.3736	0.9793	0.9314
18	20	2	2	-0.2945	0.3297	-0.0726	-0.0820	0.9921	0.9738
18	20	2	3	0.5952	-0.6662	-0.3669	-0.4143	0.0	0.0
18	20	3	3	0.0859	-0.0962	-0.1001	-0.1131	0.9685	0.8965
18	21	3	3	-0.5154	0.5759	0.0347	0.0391	0.9887	0.9628
19	16	3	3	-0.6037	0.6757	0.0588	0.0663	0.9861	0.9541
19	17	2	2	-0.3450	0.3861	-0.1231	-0.1390	0.9912	0.9709
19	17	2	3	-0.5951	0.6660	0.5310	0.5994	0.0	0.0

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
19	17	3	3	-0.0906	0.1014	-0.1078	-0.1216	0.9650	0.8853
19	18	1	1	0.2415	-0.2703	0.0	0.0	0.9958	0.9861
19	18	1	2	-0.8873	0.9930	0.0	0.0	0.0	0.0
19	18	2	2	0.1207	-0.1351	0.4310	0.4865	0.9793	0.9314
19	19	1	1	-0.4468	0.5000	0.0	0.0	0.9921	0.9737
19	19	1	2	-0.0770	0.0861	0.0	0.0	0.0	0.0
19	19	2	2	0.3166	-0.3543	-0.5677	-0.6408	0.9764	0.9220
19	20	1	1	0.2070	-0.2317	0.0	0.0	0.9963	0.9875
19	20	1	2	0.8430	-0.9435	0.0	0.0	0.0	0.0
19	20	2	2	0.1946	-0.2178	0.3334	0.3763	0.9813	0.9380
19	21	2	2	-0.2958	0.3310	-0.0737	-0.0831	0.9929	0.9763
19	21	2	3	0.5954	-0.6664	-0.3707	-0.4184	0.0	0.0
19	21	3	3	0.0817	-0.0915	-0.1005	-0.1135	0.9715	0.9062
19	22	3	3	-0.5176	0.5793	0.0352	0.0397	0.9898	0.9662
20	17	3	3	-0.6015	0.6731	0.0580	0.0655	0.9875	0.9588
20	18	2	2	-0.3437	0.3846	-0.1216	-0.1372	0.9921	0.9738
20	18	2	3	-0.5953	0.6662	0.5266	0.5942	0.0	0.0
20	18	3	3	-0.0859	0.0962	-0.1078	-0.1216	0.9685	0.8965
20	19	1	1	0.2406	-0.2692	0.0	0.0	0.9963	0.9875
20	19	1	2	-0.8863	0.9918	0.0	0.0	0.0	0.0
20	19	2	2	0.1227	-0.1374	0.4285	0.4835	0.9813	0.9380
20	20	1	1	-0.4468	0.5000	0.0	0.0	0.9929	0.9762
20	20	1	2	-0.0732	0.0819	0.0	0.0	0.0	0.0
20	20	2	2	0.3169	-0.3546	-0.5680	-0.6410	0.9786	0.9293
20	21	1	1	0.2078	-0.2326	0.0	0.0	0.9966	0.9887
20	21	1	2	0.8442	-0.9447	0.0	0.0	0.0	0.0
20	21	2	2	0.1930	-0.2159	0.3357	0.3787	0.9830	0.9437
20	22	2	2	-0.2969	0.3322	-0.0746	-0.0842	0.9935	0.9784
20	22	2	3	0.5956	-0.6665	-0.3741	-0.4221	0.0	0.0
20	22	3	3	0.0779	-0.0872	-0.1009	-0.1138	0.9741	0.9146
20	23	3	3	-0.5195	0.5814	0.0356	0.0402	0.9907	0.9692

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
3/2	1/2	1	1	0.5000	-0.5000	0.0	0.0	0.0	0.0
3/2	1/2	1	2	-0.8660	0.8660	0.0	0.0	0.0	0.0
3/2	1/2	2	2	-0.5000	0.5000	0.0	0.0	0.0	0.0
3/2	3/2	1	1	-0.4000	0.4000	0.0	0.0	0.2000	0.0
3/2	3/2	1	2	-0.7746	0.7746	0.0	0.0	0.0	0.0
3/2	3/2	2	2	0.0	0.0	0.0	0.0	-0.6000	0.0
3/2	5/2	1	1	0.1000	-0.1000	0.0	0.0	0.7483	0.0
3/2	5/2	1	2	0.5916	-0.5916	0.0	0.0	0.0	0.0
3/2	5/2	2	2	0.3571	-0.3571	0.0	0.0	-0.1069	0.0
3/2	7/2	2	2	-0.1429	0.1429	0.0	0.0	0.6547	0.0
3/2	7/2	2	3	0.4629	-0.4629	0.0	0.0	0.0	0.0
3/2	7/2	3	3	0.5000	-0.5000	0.0	0.0	-0.2182	0.0
3/2	9/2	3	3	-0.2500	0.2500	0.0	0.0	0.6055	0.0
5/2	1/2	2	2	-0.5345	0.5714	-0.6172	-0.5714	0.0	0.0
5/2	1/2	2	3	-0.3780	0.4041	1.0911	1.0102	0.0	0.0
5/2	1/2	3	3	-0.8018	0.8571	0.1543	0.1429	0.0	0.0
5/2	3/2	1	1	0.3742	-0.4000	0.0	0.0	0.7483	0.0
5/2	3/2	1	2	-0.9487	1.0142	0.0	0.0	0.0	0.0
5/2	3/2	2	2	-0.1909	0.2041	0.7054	0.6531	-0.1069	0.0
5/2	5/2	1	1	-0.4276	0.4571	0.0	0.0	0.6571	-0.1429
5/2	5/2	1	2	-0.5071	0.5421	0.0	0.0	0.0	0.0
5/2	5/2	2	2	0.1909	-0.2041	-0.3968	-0.3673	0.1000	-0.5000
5/2	7/2	1	1	0.1336	-0.1429	0.0	0.0	0.8748	0.5803
5/2	7/2	1	2	0.6944	-0.7423	0.0	0.0	0.0	0.0
5/2	7/2	2	2	0.3245	-0.3469	0.1176	0.1088	0.4082	-0.4513
5/2	9/2	2	2	-0.1909	0.2041	-0.0147	-0.0136	0.8092	0.4349
5/2	9/2	2	3	0.5297	-0.5663	-0.1019	-0.0944	0.0	0.0
5/2	9/2	3	3	0.4009	-0.4286	-0.0444	-0.0411	0.2795	-0.5140
5/2	11/2	3	3	-0.3341	0.3571	0.0070	0.0065	0.7687	0.3624
7/2	1/2	3	3	-0.8183	0.8929	0.1709	0.1753	0.0	0.0

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
7/2	3/2	2	2	-0.4676	0.5102	-0.3582	-0.3673	0.6547	0.0
7/2	3/2	2	3	-0.5051	0.5511	0.9671	0.9920	0.0	0.0
7/2	3/2	3	3	-0.5455	0.5952	-0.0190	-0.0195	-0.2182	0.0
7/2	5/2	1	1	0.3273	-0.3571	0.0	0.0	0.8748	0.5803
7/2	5/2	1	2	-0.9449	1.0310	0.0	0.0	0.0	0.0
7/2	5/2	2	2	-0.0779	0.0850	0.6367	0.6531	0.4082	-0.4513
7/2	7/2	1	1	-0.4364	0.4762	0.0	0.0	0.8095	0.3651
7/2	7/2	1	2	-0.3780	0.4124	0.0	0.0	0.0	0.0
7/2	7/2	2	2	0.2494	-0.2721	-0.4775	-0.4898	0.4667	-0.3333
7/2	9/2	1	1	0.1528	-0.1667	0.0	0.0	0.9250	0.7495
7/2	9/2	1	2	0.7416	-0.8092	0.0	0.0	0.0	0.0
7/2	9/2	2	2	0.2962	-0.3231	0.1737	0.1781	0.6367	-0.0292
7/2	11/2	2	2	-0.2192	0.2381	-0.0253	-0.0260	0.8787	0.6245
7/2	11/2	2	3	0.5563	-0.6070	-0.1614	-0.1656	0.0	0.0
7/2	11/2	3	3	0.3273	-0.3571	-0.0627	-0.0643	0.5311	-0.1990
7/2	13/2	3	3	-0.3819	0.4167	0.0121	0.0124	0.8473	0.5501
9/2	3/2	3	3	-0.7569	0.8333	0.1281	0.1364	0.6055	0.0
9/2	5/2	2	2	-0.4325	0.4762	-0.2684	-0.2857	0.8092	0.4349
9/2	5/2	2	3	-0.5455	0.6006	0.8465	0.9009	0.0	0.0
9/2	5/2	3	3	-0.4129	0.4545	-0.0660	-0.0702	0.2795	-0.5140
9/2	7/2	1	1	0.3028	-0.3333	0.0	0.0	0.9250	0.7495
9/2	7/2	1	2	-0.9354	1.0299	0.0	0.0	0.0	0.0
9/2	7/2	2	2	-0.0197	0.0216	0.5857	0.6234	0.6367	-0.0292
9/2	9/2	1	1	-0.4404	0.4848	0.0	0.0	0.8788	0.5960
9/2	9/2	1	2	-0.3015	0.3320	0.0	0.0	0.0	0.0
9/2	9/2	2	2	0.2752	-0.3030	-0.5125	-0.5455	0.6515	0.0152
9/2	11/2	1	1	0.1651	-0.1818	0.0	0.0	0.9500	0.8332
9/2	11/2	1	2	0.7687	-0.8463	0.0	0.0	0.0	0.0
9/2	11/2	2	2	0.2752	-0.3030	0.2102	0.2238	0.7551	0.2635
9/2	13/2	2	2	-0.2359	0.2597	-0.0338	-0.0360	0.9161	0.7340

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
9/2	13/2	2	3	0.5698	-0.6273	-0.2040	-0.2172	0.0	0.0
9/2	13/2	3	3	0.2752	-0.3030	-0.0735	-0.0782	0.6718	0.0816
9/2	15/2	3	3	-0.4129	0.4545	0.0161	0.0172	0.8915	0.6684
11/2	5/2	3	3	-0.7191	0.7955	0.1067	0.1157	0.7687	0.3624
11/2	7/2	2	2	-0.4109	0.4545	-0.2237	-0.2424	0.8787	0.6245
11/2	7/2	2	3	-0.5641	0.6240	0.7676	0.8320	0.0	0.0
11/2	7/2	3	3	-0.3319	0.3671	-0.0849	-0.0920	0.5311	-0.1990
11/2	9/2	1	1	0.2876	-0.3182	0.0	0.0	0.9500	0.8332
11/2	9/2	1	2	-0.9269	1.0253	0.0	0.0	0.0	0.0
11/2	9/2	2	2	0.0158	-0.0175	0.5505	0.5967	0.7551	0.2635
11/2	11/2	1	1	-0.4425	0.4895	0.0	0.0	0.9161	0.7203
11/2	11/2	1	2	-0.2509	0.2775	0.0	0.0	0.0	0.0
11/2	11/2	2	2	0.2890	-0.3197	-0.5309	-0.5754	0.7554	0.2687
11/2	13/2	1	1	0.1738	-0.1923	0.0	0.0	0.9643	0.8809
11/2	13/2	1	2	0.7862	-0.8697	0.0	0.0	0.0	0.0
11/2	13/2	2	2	0.2596	-0.2872	0.2359	0.2557	0.8240	0.4538
11/2	15/2	2	2	-0.2483	0.2747	-0.0406	-0.0440	0.9385	0.8022
11/2	15/2	2	3	0.5776	-0.6389	-0.2358	-0.2556	0.0	0.0
11/2	15/2	3	3	0.2371	-0.2622	-0.0804	-0.0871	0.7578	0.2874
11/2	17/2	3	3	-0.4346	0.4808	0.0194	0.0210	0.9189	0.7465
13/2	7/2	3	3	-0.6934	0.7692	0.0940	0.1030	0.8473	0.5501
13/2	9/2	2	2	-0.3962	0.4396	-0.1970	-0.2158	0.9161	0.7340
13/2	9/2	2	3	-0.5742	0.6370	0.7138	0.7818	0.0	0.0
13/2	9/2	3	3	-0.2773	0.3077	-0.0940	-0.1030	0.6718	0.0816
13/2	11/2	1	1	0.2774	-0.3077	0.0	0.0	0.9643	0.8809
13/2	11/2	1	2	-0.9199	1.0205	0.0	0.0	0.0	0.0
13/2	11/2	2	2	0.0396	-0.0440	0.5254	0.5754	0.8240	0.4538
13/2	13/2	1	1	-0.4438	0.4923	0.0	0.0	0.9385	0.7949
13/2	13/2	1	2	-0.2148	0.2383	0.0	0.0	0.0	0.0
13/2	13/2	2	2	0.2972	-0.3297	-0.5418	-0.5934	0.8192	0.4423

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
13/2	15/2	1	1	0.1803	-0.2000	0.0	0.0	0.9732	0.9107
13/2	15/2	1	2	0.7984	-0.8858	0.0	0.0	0.0	0.0
13/2	15/2	2	2	0.2476	-0.2747	0.2550	0.2793	0.8675	0.5810
13/2	17/2	2	2	-0.2575	0.2857	-0.0460	-0.0504	0.9529	0.8475
13/2	17/2	2	3	0.5825	-0.6462	-0.2603	-0.2851	0.0	0.0
13/2	17/2	3	3	0.2080	-0.2308	-0.0851	-0.0932	0.8141	0.4357
13/2	19/2	3	3	-0.4507	0.5000	0.0220	0.0241	0.9371	0.8004
15/2	9/2	3	3	-0.6748	0.7500	0.0856	0.0944	0.8915	0.6684
15/2	11/2	2	2	-0.3856	0.4286	-0.1794	-0.1978	0.9385	0.8022
15/2	11/2	2	3	-0.5803	0.6450	0.6751	0.7442	0.0	0.0
15/2	11/2	3	3	-0.2382	0.2647	-0.0991	-0.1092	0.7578	0.2874
15/2	13/2	1	1	0.2699	-0.3000	0.0	0.0	0.9732	0.9107
15/2	13/2	1	2	-0.9142	1.0160	0.0	0.0	0.0	0.0
15/2	13/2	2	2	0.0567	-0.0630	0.5066	0.5585	0.8675	0.5810
15/2	15/2	1	1	-0.4446	0.4941	0.0	0.0	0.9530	0.8431
15/2	15/2	1	2	-0.1879	0.2088	0.0	0.0	0.0	0.0
15/2	15/2	2	2	0.3024	-0.3361	-0.5488	-0.6050	0.8611	0.5630
15/2	17/2	1	1	0.1852	-0.2059	0.0	0.0	0.9792	0.9305
15/2	17/2	1	2	0.8074	-0.8974	0.0	0.0	0.0	0.0
15/2	17/2	2	2	0.2382	-0.2647	0.2696	0.2972	0.8967	0.6693
15/2	19/2	2	2	-0.2646	0.2941	-0.0506	-0.0557	0.9628	0.8788
15/2	19/2	2	3	0.5858	-0.6511	-0.2797	-0.3084	0.0	0.0
15/2	19/2	3	3	0.1852	-0.2059	-0.0885	-0.0975	0.8528	0.5440
15/2	21/2	3	3	-0.4631	0.5147	0.0241	0.0266	0.9497	0.8389
17/2	11/2	3	3	-0.6607	0.7353	0.0797	0.0882	0.9189	0.7465
17/2	13/2	2	2	-0.3776	0.4202	-0.1670	-0.1849	0.9529	0.8475
17/2	13/2	2	3	-0.5843	0.6502	0.6460	0.7152	0.0	0.0
17/2	13/2	3	3	-0.2087	0.2322	-0.1021	-0.1130	0.8141	0.4357
17/2	15/2	1	1	0.2643	-0.2941	0.0	0.0	0.9792	0.9305
17/2	15/2	1	2	-0.9095	1.0121	0.0	0.0	0.0	0.0

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
17/2	15/2	2	2	0.0696	-0.0774	0.4921	0.5449	0.8967	0.6693
17/2	17/2	1	1	-0.4451	0.4954	0.0	0.0	0.9628	0.8762
17/2	17/2	1	2	-0.1669	0.1858	0.0	0.0	0.0	0.0
17/2	17/2	2	2	0.3060	-0.3406	-0.5536	-0.6130	0.8899	0.6494
17/2	19/2	1	1	0.1892	-0.2105	0.0	0.0	0.9833	0.9444
17/2	19/2	1	2	0.8143	-0.9062	0.0	0.0	0.0	0.0
17/2	19/2	2	2	0.2305	-0.2565	0.2812	0.3114	0.9172	0.7328
17/2	21/2	2	2	-0.2703	0.3007	-0.0543	-0.0601	0.9699	0.9015
17/2	21/2	2	3	0.5881	-0.6545	-0.2955	-0.3272	0.0	0.0
17/2	21/2	3	3	0.1669	-0.1858	-0.0910	-0.1008	0.8806	0.6248
17/2	23/2	3	3	-0.4730	0.5263	0.0259	0.0287	0.9589	0.8673
19/2	13/2	3	3	-0.6497	0.7237	0.0753	0.0836	0.9371	0.8004
19/2	15/2	2	2	-0.3713	0.4135	-0.1577	-0.1751	0.9628	0.8788
19/2	15/2	2	3	-0.5870	0.6538	0.6233	0.6923	0.0	0.0
19/2	15/2	3	3	-0.1856	0.2068	-0.1039	-0.1154	0.8528	0.5440
19/2	17/2	1	1	0.2599	-0.2895	0.0	0.0	0.9833	0.9444
19/2	17/2	1	2	-0.9056	1.0087	0.0	0.0	0.0	0.0
19/2	17/2	2	2	0.0796	-0.0886	0.4806	0.5338	0.9172	0.7328
19/2	19/2	1	1	-0.4455	0.4962	0.0	0.0	0.9699	0.8997
19/2	19/2	1	2	-0.1502	0.1673	0.0	0.0	0.0	0.0
19/2	19/2	2	2	0.3086	-0.3437	-0.5570	-0.6187	0.9107	0.7129
19/2	21/2	1	1	0.1924	-0.2143	0.0	0.0	0.9864	0.9545
19/2	21/2	1	2	0.8198	-0.9131	0.0	0.0	0.0	0.0
19/2	21/2	2	2	0.2242	-0.2497	0.2906	0.3228	0.9322	0.7798
19/2	23/2	2	2	-0.2748	0.3061	-0.0575	-0.0639	0.9752	0.9184
19/2	23/2	2	3	0.5898	-0.6569	-0.3086	-0.3427	0.0	0.0
19/2	23/2	3	3	0.1519	-0.1692	-0.0930	-0.1032	0.9012	0.6864
19/2	25/2	3	3	-0.4810	0.5357	0.0275	0.0305	0.9658	0.8889
21/2	15/2	3	3	-0.6409	0.7143	0.0718	0.0800	0.9497	0.8389
21/2	17/2	2	2	-0.3662	0.4082	-0.1505	-0.1676	0.9699	0.9015

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
21/2	17/2	2	3	-0.5890	0.6565	0.6052	0.6738	0.0	0.0
21/2	17/2	3	3	-0.1672	0.1863	-0.1052	-0.1171	0.8806	0.6248
21/2	19/2	1	1	0.2563	-0.2857	0.0	0.0	0.9864	0.9545
21/2	19/2	1	2	-0.9024	1.0057	0.0	0.0	0.0	0.0
21/2	19/2	2	2	0.0876	-0.0976	0.4712	0.5245	0.9322	0.7798
21/2	21/2	1	1	-0.4458	0.4969	0.0	0.0	0.9752	0.9172
21/2	21/2	1	2	-0.1365	0.1521	0.0	0.0	0.0	0.0
21/2	21/2	2	2	0.3105	-0.3460	-0.5595	-0.6229	0.9261	0.7609
21/2	23/2	1	1	0.1950	-0.2174	0.0	0.0	0.9886	0.9621
21/2	23/2	1	2	0.8242	-0.9186	0.0	0.0	0.0	0.0
21/2	23/2	2	2	0.2189	-0.2440	0.2984	0.3322	0.9434	0.8155
21/2	25/2	2	2	-0.2786	0.3106	-0.0603	-0.0671	0.9791	0.9313
21/2	25/2	2	3	0.5911	-0.6588	-0.3196	-0.3558	0.0	0.0
21/2	25/2	3	3	0.1393	-0.1553	-0.0945	-0.1052	0.9170	0.7342
21/2	27/2	3	3	-0.4876	0.5435	0.0288	0.0320	0.9711	0.9057
23/2	17/2	3	3	-0.6336	0.7065	0.0691	0.0771	0.9589	0.8673
23/2	19/2	2	2	-0.3620	0.4037	-0.1448	-0.1615	0.9752	0.9184
23/2	19/2	2	3	-0.5905	0.6585	0.5905	0.6585	0.0	0.0
23/2	19/2	3	3	-0.1521	0.1696	-0.1060	-0.1182	0.9012	0.6864
23/2	21/2	1	1	0.2534	-0.2826	0.0	0.0	0.9886	0.9621
23/2	21/2	1	2	-0.8996	1.0032	0.0	0.0	0.0	0.0
23/2	21/2	2	2	0.0941	-0.1050	0.4634	0.5168	0.9434	0.8155
23/2	23/2	1	1	-0.4460	0.4974	0.0	0.0	0.9791	0.9304
23/2	23/2	1	2	-0.1251	0.1395	0.0	0.0	0.0	0.0
23/2	23/2	2	2	0.3119	-0.3478	-0.5615	-0.6261	0.9378	0.7970
23/2	25/2	1	1	0.1973	-0.2200	0.0	0.0	0.9904	0.9680
23/2	25/2	1	2	0.8279	-0.9232	0.0	0.0	0.0	0.0
23/2	25/2	2	2	0.2144	-0.2391	0.3050	0.3401	0.9521	0.8433
23/2	27/2	2	2	-0.2818	0.3143	-0.0626	-0.0698	0.9822	0.9414
23/2	27/2	2	3	0.5921	-0.6602	-0.3289	-0.3668	0.0	0.0

JT	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
23/2	27/2	3	3	0.1287	-0.1435	-0.0957	-0.1068	0.9292	0.7720
23/2	29/2	3	3	-0.4932	0.5500	0.0299	0.0333	0.9752	0.9190
25/2	19/2	3	3	-0.6275	0.7000	0.0669	0.0747	0.9658	0.8889
25/2	21/2	2	2	-0.3586	0.4000	-0.1402	-0.1565	0.9791	0.9313
25/2	21/2	2	3	-0.5916	0.6600	0.5782	0.6456	0.0	0.0
25/2	21/2	3	3	-0.1394	0.1556	-0.1065	-0.1190	0.9170	0.7342
25/2	23/2	1	1	0.2510	-0.2800	0.0	0.0	0.9904	0.9680
25/2	23/2	1	2	-0.8972	1.0009	0.0	0.0	0.0	0.0
25/2	23/2	2	2	0.0996	-0.1111	0.4569	0.5101	0.9521	0.8433
25/2	25/2	1	1	-0.4462	0.4978	0.0	0.0	0.9822	0.9408
25/2	25/2	1	2	-0.1155	0.1288	0.0	0.0	0.0	0.0
25/2	25/2	2	2	0.3130	-0.3492	-0.5629	-0.6286	0.9470	0.8270
25/2	27/2	1	1	0.1992	-0.2222	0.0	0.0	0.9918	0.9725
25/2	27/2	1	2	0.8310	-0.9270	0.0	0.0	0.0	0.0
25/2	27/2	2	2	0.2106	-0.2349	0.3106	0.3468	0.9589	0.8652
25/2	29/2	2	2	-0.2846	0.3175	-0.0647	-0.0722	0.9847	0.9494
25/2	29/2	2	3	0.5929	-0.6614	-0.3370	-0.3763	0.0	0.0
25/2	29/2	3	3	0.1195	-0.1333	-0.0968	-0.1080	0.9389	0.8024
25/2	31/2	3	3	-0.4980	0.5556	0.0309	0.0345	0.9785	0.9296
27/2	21/2	3	3	-0.6223	0.6944	0.0651	0.0727	0.9711	0.9057
27/2	23/2	2	2	-0.3556	0.3968	-0.1363	-0.1524	0.9822	0.9414
27/2	23/2	2	3	-0.5925	0.6612	0.5678	0.6347	0.0	0.0
27/2	23/2	3	3	-0.1288	0.1437	-0.1069	-0.1195	0.9292	0.7720
27/2	25/2	1	1	0.2489	-0.2778	0.0	0.0	0.9918	0.9725
27/2	25/2	1	2	-0.8951	0.9989	0.0	0.0	0.0	0.0
27/2	25/2	2	2	0.1042	-0.1163	0.4513	0.5044	0.9589	0.8652
27/2	27/2	1	1	-0.4464	0.4981	0.0	0.0	0.9847	0.9489
27/2	27/2	1	2	-0.1072	0.1196	0.0	0.0	0.0	0.0
27/2	27/2	2	2	0.3139	-0.3503	-0.5641	-0.6305	0.9543	0.8503
27/2	29/2	1	1	0.2009	-0.2241	0.0	0.0	0.9929	0.9762

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
27/2	29/2	1	2	0.8336	-0.9302	0.0	0.0	0.0	0.0
27/2	29/2	2	2	0.2072	-0.2313	0.3154	0.3526	0.9644	0.8829
27/2	31/2	2	2	-0.2869	0.3202	-0.0665	-0.0744	0.9867	0.9559
27/2	31/2	2	3	0.5935	-0.6623	-0.3440	-0.3845	0.0	0.0
27/2	31/2	3	3	0.1116	-0.1245	-0.0976	-0.1091	0.9468	0.8271
27/2	33/2	3	3	-0.5022	0.5603	0.0318	0.0355	0.9812	0.9383
29/2	23/2	3	3	-0.6179	0.6897	0.0635	0.0711	0.9752	0.9190
29/2	25/2	2	2	-0.3531	0.3941	-0.1331	-0.1489	0.9847	0.9494
29/2	25/2	2	3	-0.5932	0.6621	0.5590	0.6253	0.0	0.0
29/2	25/2	3	3	-0.1196	0.1335	-0.1072	-0.1200	0.9389	0.8024
29/2	27/2	1	1	0.2472	-0.2759	0.0	0.0	0.9929	0.9762
29/2	27/2	1	2	-0.8933	0.9971	0.0	0.0	0.0	0.0
29/2	27/2	2	2	0.1082	-0.1208	0.4465	0.4995	0.9644	0.8829
29/2	29/2	1	1	-0.4465	0.4983	0.0	0.0	0.9866	0.9555
29/2	29/2	1	2	-0.1001	0.1117	0.0	0.0	0.0	0.0
29/2	29/2	2	2	0.3146	-0.3512	-0.5651	-0.6321	0.9601	0.8692
29/2	31/2	1	1	0.2023	-0.2258	0.0	0.0	0.9938	0.9792
29/2	31/2	1	2	0.8358	-0.9329	0.0	0.0	0.0	0.0
29/2	31/2	2	2	0.2043	-0.2280	0.3196	0.3576	0.9688	0.8973
29/2	33/2	2	2	-0.2890	0.3226	-0.0682	-0.0762	0.9883	0.9612
29/2	33/2	2	3	0.5940	-0.6630	-0.3502	-0.3918	0.0	0.0
29/2	33/2	3	3	0.1046	-0.1168	-0.0983	-0.1100	0.9532	0.8475
29/2	35/2	3	3	-0.5058	0.5645	0.0325	0.0364	0.9834	0.9455
31/2	25/2	3	3	-0.6140	0.6855	0.0622	0.0696	0.9785	0.9296
31/2	27/2	2	2	-0.3509	0.3917	-0.1303	-0.1459	0.9867	0.9559
31/2	27/2	2	3	-0.5937	0.6629	0.5513	0.6171	0.0	0.0
31/2	27/2	3	3	-0.1116	0.1246	-0.1074	-0.1203	0.9468	0.8271
31/2	29/2	1	1	0.2456	-0.2742	0.0	0.0	0.9938	0.9792
31/2	29/2	1	2	-0.8917	0.9955	0.0	0.0	0.0	0.0
31/2	29/2	2	2	0.1116	-0.1246	0.4423	0.4951	0.9688	0.8973

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
31/2	33/2	2	2	0.2017	-0.2252	0.3233	0.3620	0.9725	0.9092
31/2	35/2	2	2	-0.2908	0.3247	-0.0696	-0.0779	0.9896	0.9656
31/2	35/2	2	3	0.5944	-0.6636	-0.3557	-0.3982	0.0	0.0
31/2	35/2	3	3	0.0985	-0.1100	-0.0989	-0.1108	0.9586	0.8646
31/2	37/2	3	3	-0.5089	0.5682	0.0332	0.0372	0.9853	0.9515
31/2	31/2	1	1	-0.4466	0.4985	0.0	0.0	0.9883	0.9609
31/2	31/2	1	2	-0.0938	0.1047	0.0	0.0	0.0	0.0
31/2	31/2	2	2	0.3152	-0.3519	-0.5658	-0.6334	0.9649	0.8848
31/2	33/2	1	1	0.2036	-0.2273	0.0	0.0	0.9945	0.9816
31/2	33/2	1	2	0.8378	-0.9353	0.0	0.0	0.0	0.0
33/2	27/2	3	3	-0.6106	0.6818	0.0610	0.0684	0.9812	0.9383
33/2	29/2	2	2	-0.3489	0.3896	-0.1279	-0.1433	0.9883	0.9612
33/2	29/2	2	3	-0.5942	0.6635	0.5446	0.6100	0.0	0.0
33/2	29/2	3	3	-0.1047	0.1169	-0.1076	-0.1205	0.9532	0.8475
33/2	31/2	1	1	0.2443	-0.2727	0.0	0.0	0.9945	0.9816
33/2	31/2	1	2	-0.8903	0.9941	0.0	0.0	0.0	0.0
33/2	31/2	2	2	0.1146	-0.1280	0.4386	0.4912	0.9725	0.9092
33/2	33/2	1	1	-0.4466	0.4987	0.0	0.0	0.9896	0.9654
33/2	33/2	1	2	-0.0883	0.0986	0.0	0.0	0.0	0.0
33/2	33/2	2	2	0.3157	-0.3525	-0.5665	-0.6345	0.9690	0.8977
33/2	35/2	1	1	0.2047	-0.2286	0.0	0.0	0.9951	0.9837
33/2	35/2	1	2	0.8395	-0.9374	0.0	0.0	0.0	0.0
33/2	35/2	2	2	0.1994	-0.2226	0.3266	0.3658	0.9755	0.9192
33/2	37/2	2	2	-0.2924	0.3265	-0.0709	-0.0794	0.9907	0.9693
33/2	37/2	2	3	0.5948	-0.6641	-0.3605	-0.4038	0.0	0.0
33/2	37/2	3	3	0.0930	-0.1039	-0.0995	-0.1114	0.9630	0.8789
33/2	39/2	3	3	-0.5118	0.5714	0.0338	0.0379	0.9868	0.9565
35/2	29/2	3	3	-0.6076	0.6786	0.0601	0.0673	0.9834	0.9455
35/2	31/2	2	2	-0.3472	0.3878	-0.1258	-0.1410	0.9896	0.9656
35/2	31/2	2	3	-0.5946	0.6640	0.5387	0.6036	0.0	0.0

J1	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
35/2	31/2	3	3	-0.0985	0.1100	-0.1077	-0.1206	0.9586	0.8646
35/2	33/2	1	1	0.2431	-0.2714	0.0	0.0	0.9951	0.9837
35/2	33/2	1	2	-0.8890	0.9928	0.0	0.0	0.0	0.0
35/2	33/2	2	2	0.1173	-0.1310	0.4353	0.4878	0.9755	0.9192
35/2	35/2	1	1	-0.4467	0.4988	0.0	0.0	0.9907	0.9691
35/2	35/2	1	2	-0.0834	0.0931	0.0	0.0	0.0	0.0
35/2	35/2	2	2	0.3161	-0.3530	-0.5670	-0.6354	0.9723	0.9086
35/2	37/2	1	1	0.2057	-0.2297	0.0	0.0	0.9956	0.9854
35/2	37/2	1	2	0.8410	-0.9392	0.0	0.0	0.0	0.0
35/2	37/2	2	2	0.1973	-0.2204	0.3295	0.3693	0.9781	0.9276
35/2	39/2	2	2	-0.2939	0.3282	-0.0721	-0.0808	0.9917	0.9724
35/2	39/2	2	3	0.5951	-0.6645	-0.3649	-0.4089	0.0	0.0
35/2	39/2	3	3	0.0882	-0.0985	-0.0999	-0.1120	0.9668	0.8911
35/2	41/2	3	3	-0.5143	0.5743	0.0344	0.0386	0.9881	0.9608
37/2	31/2	3	3	-0.6050	0.6757	0.0592	0.0663	0.9853	0.9515
37/2	33/2	2	2	-0.3457	0.3861	-0.1240	-0.1390	0.9907	0.9693
37/2	33/2	2	3	-0.5949	0.6645	0.5334	0.5980	0.0	0.0
37/2	33/2	3	3	-0.0931	0.1040	-0.1077	-0.1208	0.9630	0.8789
37/2	35/2	1	1	0.2420	-0.2703	0.0	0.0	0.9956	0.9854
37/2	35/2	1	2	-0.8879	0.9916	0.0	0.0	0.0	0.0
37/2	35/2	2	2	0.1197	-0.1337	0.4324	0.4847	0.9781	0.9276
37/2	37/2	1	1	-0.4467	0.4990	0.0	0.0	0.9917	0.9723
37/2	37/2	1	2	-0.0790	0.0882	0.0	0.0	0.0	0.0
37/2	37/2	2	2	0.3164	-0.3534	-0.5675	-0.6362	0.9751	0.9179
37/2	39/2	1	1	0.2066	-0.2308	0.0	0.0	0.9961	0.9869
37/2	39/2	1	2	0.8424	-0.9409	0.0	0.0	0.0	0.0
37/2	39/2	2	2	0.1955	-0.2183	0.3322	0.3724	0.9803	0.9348
37/2	41/2	2	2	-0.2952	0.3297	-0.0732	-0.0820	0.9925	0.9751
37/2	41/2	2	3	0.5953	-0.6649	-0.3689	-0.4135	0.0	0.0
37/2	41/2	3	3	0.0838	-0.0936	-0.1003	-0.1125	0.9700	0.9016

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
37/2	43/2	3	3	-0.5166	0.5769	0.0349	0.0391	0.9893	0.9646
39/2	33/2	3	3	-0.6026	0.6731	0.0584	0.0655	0.9868	0.9565
39/2	35/2	2	2	-0.3443	0.3846	-0.1224	-0.1372	0.9917	0.9724
39/2	35/2	2	3	-0.5952	0.6648	0.5287	0.5930	0.0	0.0
39/2	35/2	3	3	-0.0882	0.0985	-0.1078	-0.1209	0.9668	0.8911
39/2	37/2	1	1	0.2410	-0.2692	0.0	0.0	0.9961	0.9869
39/2	37/2	1	2	-0.8868	0.9906	0.0	0.0	0.0	0.0
39/2	37/2	2	2	0.1218	-0.1360	0.4297	0.4819	0.9803	0.9348
39/2	39/2	1	1	-0.4468	0.4991	0.0	0.0	0.9925	0.9750
39/2	39/2	1	2	-0.0750	0.0838	0.0	0.0	0.0	0.0
39/2	39/2	2	2	0.3167	-0.3538	-0.5679	-0.6368	0.9775	0.9258
39/2	41/2	1	1	0.2074	-0.2317	0.0	0.0	0.9964	0.9881
39/2	41/2	1	2	0.8436	-0.9423	0.0	0.0	0.0	0.0
39/2	41/2	2	2	0.1938	-0.2164	0.3345	0.3752	0.9822	0.9410
39/2	43/2	2	2	-0.2963	0.3310	-0.0741	-0.0831	0.9932	0.9774
39/2	43/2	2	3	0.5955	-0.6652	-0.3724	-0.4177	0.0	0.0
39/2	43/2	3	3	0.0798	-0.0891	-0.1007	-0.1129	0.9728	0.9106
39/2	45/2	3	3	-0.5186	0.5793	0.0354	0.0397	0.9903	0.9677
41/2	35/2	3	3	-0.6004	0.6707	0.0577	0.0647	0.9881	0.9608
41/2	37/2	2	2	-0.3431	0.3833	-0.1209	-0.1356	0.9925	0.9751
41/2	37/2	2	3	-0.5954	0.6651	0.5245	0.5884	0.0	0.0
41/2	37/2	3	3	-0.0838	0.0936	-0.1078	-0.1209	0.9700	0.9016
41/2	39/2	1	1	0.2402	-0.2683	0.0	0.0	0.9964	0.9881
41/2	39/2	1	2	-0.8859	0.9896	0.0	0.0	0.0	0.0
41/2	39/2	2	2	0.1237	-0.1382	0.4274	0.4794	0.9822	0.9410

JI	JF	L1	L2	F2	B2F2	F4	B4F4	U2	U4
41/2	41/2	1	1	-0.4468	0.4991	0.0	0.0	0.9932	0.9773
41/2	41/2	1	2	-0.0715	0.0798	0.0	0.0	0.0	0.0
41/2	41/2	2	2	0.3170	-0.3541	-0.5682	-0.6374	0.9796	0.9326
41/2	43/2	1	1	0.2082	-0.2326	0.0	0.0	0.9968	0.9892
41/2	43/2	1	2	0.8447	-0.9436	0.0	0.0	0.0	0.0
41/2	43/2	2	2	0.1922	-0.2147	0.3367	0.3777	0.9838	0.9463
41/2	45/2	2	2	-0.2974	0.3322	-0.0750	-0.0842	0.9938	0.9794
41/2	45/2	2	3	0.5957	-0.6655	-0.3757	-0.4215	0.0	0.0
41/2	45/2	3	3	0.0762	-0.0851	-0.1010	-0.1133	0.9752	0.9184
41/2	47/2	3	3	-0.5205	0.5814	0.0358	0.0402	0.9911	0.9705